

# KHUSHI GUPTA

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## Profile Summary

- B.Tech CSE (2023–2027) student with hands-on experience in software design, development, and testing, including automation testing and Python-based systems.
- Strong foundation in OOP, data structures & algorithms, SDLC, and debugging, with experience applying engineering principles in Agile, collaborative environments.

## Education

### UPES, Dehradun

Bachelor of Technology in Computer Science and Engineering - CGPA: 8.65

August 2023 – Present

Dehradun, Uttarakhand

### St. Xavier's Sr. Sec. School, Delhi

Class XII (CBSE) – Science Stream with Computer Science — Percentage: 90%

April 2019 – May 2021

Delhi

## Technical Skills

**Languages:** Python, C++, SQL

**Libraries/Frameworks:** Pandas, NumPy, Scikit-learn, XGBoost, HuggingFace

**Tools:** VS Code, Jupyter, Google Colab, Github

**Concepts:** OOP, DSA, Secure Data Handling, Privacy-Preserving Analytics, SDLC, Debugging, Agile Basics

## Work Experience

### Indian Institute of Technology (IIT), Kanpur

February 2025 – March 2025

Research Intern

Kanpur, Uttar Pradesh

- Built a RAG-based sentiment analyzer integrating LLaMA + semantic search, improving retrieval relevance by 20%.
- Fine-tuned small LLMs for emotion classification on 1000+ journal entries, boosting class consistency across 4+ emotion types.

## Projects

### Interpretable Crime Risk Assessment System | Python, Pandas, Scikit-learn, XGBoost, Geospatial Analysis

January 2026 - Present

- Engineered spatial hotspot detection using grid-based aggregation on a multi-million record Chicago crime dataset to generate normalized hotspot scores.
- Created High/Medium/Low crime risk labels from spatial intensity patterns to enable supervised learning for XGBoost-based risk prediction.
- Implemented privacy-aware data governance with coordinate masking (100m precision) and role-based dataset views (public analytics vs secure model training).

### RAG Journal Entry Sentiment Analyzer | Python, Hugging Face, LLaMA

February 2025 - March 2025

- Engineered a 4-stage RAG pipeline (preprocess → retrieval → classify → generate) for journaling insights.
- Reduced reflection-generation time by 15 seconds and improved accuracy by 20% using Sentence-BERT + FLAN-T5.
- Integrated a distilroberta-based classifier to label emotions across 1,000+ user & historical entries.
- Automated an end-to-end NLP workflow consisting of 8+ modules (cleaning, tokenization, embedding, indexing).

## Leadership and Achievements

- Scholar, SheCodes** – Selected for global coding program supporting women in tech.
- Scholar, SheFi Season 14** – Chosen for Web3 and fintech-focused cohort empowering women leaders in blockchain.
- Round 2 Qualifier, EY Techathon 6.0** – Selected among the top 2,500 teams out of 1,83,261 registrants; architected an AI-powered NBFC loan-approval chatbot with a gamified CIBIL score tracker.
- Participant, UPES CSI Hackathon 8.0** – Architected *EcoSight*, a React-based geospatial visualization platform to model clean-energy potential and long-term environmental impact.
- President, Aks** – Led 80+ members; managed club operations, secured sponsorships, and executed large-scale performances.
- Sub-Head, Avishkarnam** – Managed 15+ writers; drafted MoUs and outreach content for club events.